

PROJECT TITLE

Blockchain-Assisted Distributed AES Key Lifecycle Management Framework for Secure Healthcare Internet of Things

BACKGROUND

Healthcare Internet of Things (IoT) ecosystems continuously generate and exchange highly sensitive medical information among wearable devices, sensors, cloud servers, physicians, and healthcare institutions. While AES provides efficient encryption for protecting patient data, centralized key management introduces significant vulnerabilities including single points of failure, insider threats, unauthorized modifications, and poor auditability.

Blockchain technology offers decentralized trust, immutability, and transparent transaction logging, making it an attractive solution for distributed cryptographic key management.

PROBLEM STATEMENT

The objective of this project is to develop a **Blockchain-based Distributed AES Key Lifecycle Management Framework** capable of securely managing cryptographic keys throughout their entire lifecycle within healthcare IoT ecosystems.

The proposed framework shall automate secure AES key generation, distribution, storage, periodic rotation, revocation, archival, and emergency recovery through blockchain-enabled smart contracts. Patient records shall remain encrypted using AES-256, while blockchain shall securely maintain cryptographic metadata, audit logs, authorization policies, and key lifecycle events without exposing confidential medical information.

The framework shall provide complete traceability of key operations while maintaining compliance with healthcare security requirements.

TANGIBLE DELIVERABLES

- Healthcare Encryption Platform
- Blockchain Network
- Smart Contract-based Key Management System
- AES Encryption Module
- Secure Key Rotation Engine
- Audit and Compliance Dashboard
- Security Evaluation Framework

RESEARCH OBJECTIVES

- Eliminate centralized key management.
- Improve cryptographic transparency and accountability.
- Automate secure key lifecycle operations.
- Evaluate blockchain overhead.
- Improve resilience against insider attacks.

RESEARCH NOVELTIES

- Blockchain-governed AES lifecycle management.
- Smart contract-enabled cryptographic automation.
- Immutable cryptographic auditing.
- Distributed healthcare key governance.
- Integrated emergency key recovery protocol.

PUBLICATION SCOPE

Potential publication venues include:

- IEEE Internet of Things Journal
- IEEE Transactions on Network Science
- Journal of Network and Computer Applications
- IEEE HealthCom
- Springer LNNS
- IEEE Access